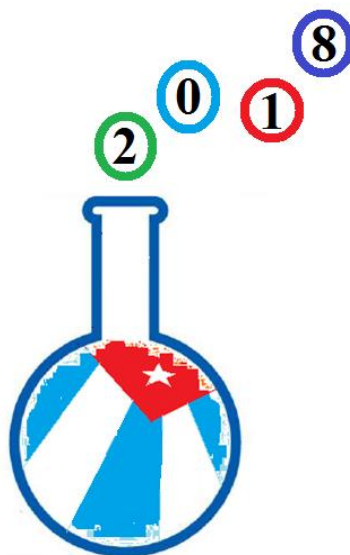


PREPARATORY PROBLEMS



CUBA NATIONAL CHEMISTRY CONTEST 2018



Ministerio de Educación
República de Cuba



PREPARATORY PROBLEMS

CUBA

National Chemistry Contest 2018

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Free distribution

Preface

To the olympists

The present material has been created for the Cuban students and mentors participating in the Cuban National Chemistry Contest of 2018, corresponding to the course 2017-2018. The aim is not to substitute already available materials but to complement and, to some extent, guide the last stage of the training after the Provincial Contest.

These Preparatory Problems were carefully selected to exercise the topics of particular difficulties, or *Advanced topics*, that are explicitly declared. These topics are the most problematic contents within the syllabus (see Annex 3) that will be evaluated in the next edition of the olympiad. This does not mean that other program topics will be excluded from the exam. I would like to let you know that the degree of difficulty and the extension of the problems are, in the majority of cases, superior to the questions in the next national exam, which corresponds with the pedagogical objective of this booklet. The answers will be explained during the National Seminar for Mentor Teachers held in Havana next October.

The booklet also includes annexes with the results of the Cuban participants in the most recent national and international events. In the future, we'll be working on elaborating an updated contest program and developing the Preparatory Problems with the contribution of mentor teachers from all over the country.

Acknowledgements

To the Ministry of Education of the Republic of Cuba to facilitate the distribution of this material. To the Faculty of Chemistry for supporting the training and laboratory practice. Likewise, I congratulate national and provincial mentors for contributing to the scientific education of thousands of students.

The author

Havana, September 2017

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CONSTANTS, CONVERSIONS AND EQUATIONS

Avogadro's Constant	$N_A = 6,0221 \cdot 10^{23} \text{ mol}^{-1}$	Ideal gas equation	$p \cdot V = n \cdot R \cdot T$
Gasses' constant	$R = 8,3145 \text{ J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	Thermodynamics First Principle	$Q = \Delta E + W$
Cero in the Celsius scale	273,15 K	Gibbs' energy	$\Delta G = \Delta H - T \cdot \Delta S$
Faraday's constant	$F = 96485 \text{ C} \cdot \text{mol}^{-1}$	Nernst's equation	$E = E^\circ - \frac{R \cdot T}{z \cdot F} \cdot \ln \frac{c_{\text{red}}}{c_{\text{ox}}}$
Water ionic product at 25 °C	$K_W = 1,00 \cdot 10^{-14}$	Arrhenius' equation	$k = A \cdot \exp \left(-\frac{E_A}{R \cdot T} \right)$
Standard conditions	$p^\circ = 100 \text{ kPa}$ $c^\circ = 1 \text{ mol} \cdot \text{L}^{-1}$		$\Delta G^\circ = -R \cdot T \cdot \ln K^\circ = -z \cdot F \cdot \Delta E^\circ$
$1 \text{ J} = 1 \text{ C} \cdot 1 \text{ V} = 1 \text{ kPa} \cdot 1 \text{ L} = 1 \text{ W} \cdot 1 \text{ s}$		$1 \text{ C} = 1 \text{ A} \cdot 1 \text{ s}$	

PERIODIC TABLE OF THE ELEMENTS

1 1A																	18 8A
1 H 1.008	2 2A											13 3A	14 4A	15 5A	16 6A	17 7A	2 He 4.003
3 Li 6.941	4 Be 9.012											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.31	3 3B	4 4B	5 5B	6 6B	7 7B	8 8B	9 8B	10 8B	11 1B	12 2B	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.88	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	31 Ga 69.72	32 Ge 72.61	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	57 La 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.8	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra (226)	89 Ac (227)	104 Rf (261)	105 Db (262)	106 Sg (263)	107 Bh (262)	108 Hs (265)	109 Mt (266)	110 Ds (269)	111 Rg (272)	112 Uub (277)		114 Uuq (277)		116 Uuh (277)		118 Uuo (277)
58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0				
90 Th 232.0	91 Pa 231.0	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)				

Proposed Tasks

Advanced topics

Common exam

Kinetics: Analysis of complex reaction mechanism by the Steady State approximation for intermediates.

Stoichiometry: Calculations involving transformation relations, sample compositions and empirical formulae of inorganic compounds.

Inorganic: Chemical properties of the representative and transition elements, including the identification of ions/substances.

10th grade

Thermochemistry: Calculations of ΔH using thermodynamic cycles.

General: Representation of molecules and description of covalent bonding by Lewis Formula, MO theory, VSREPR model, resonance theory and hybridization.

11th grade

Equilibrium: Relation between the thermodynamic magnitudes (ΔH , ΔS , ΔG , ΔE) and the equilibrium constant; applications in the gaseous phase equilibrium and in the redox process; Latimer diagrams of standard electrode potentials.

Ionic equilibrium: Calculation of PH in complex systems. Acid-base titrations, titration curve and acid-base indicators.

12th grade

Equilibrium: relation between the thermodynamic magnitudes (ΔH , ΔS , ΔG , ΔE) and the equilibrium constant; applications in redox processes; Latimer diagrams of standard electrode potentials

Organic: Chemical reactions that involve aldehyde and ketones.

Problem 1. A sulfide mixture ¹

4,00 g of a mixture containing octasulfur, zinc sulfide and iron(II) disulfide (FeS_2) were treated with excess oxygen. As result, all the sulfur in the sample was transformed into 1,90 L of sulfur dioxide, measured at 25 °C and 101,325 kPa. The resultant solid residue was dissolved in sulfuric acid. After heating at high temperature and concentrating, 4,66 g of a mixture of zinc sulfate and iron(III) sulfate were obtained.

1.1. Write the equations corresponding to the chemical reactions that happened.

1.2. Find the composition (in mass percentage) of the initial mixture.

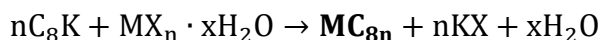
$\text{M/g}\cdot\text{mol}^{-1}$: $\text{S}_8 = 256,56$; $\text{ZnS} = 97,48$; $\text{FeS}_2 = 119,99$; $\text{ZnSO}_4 = 161,48$; $\text{Fe}_2(\text{SO}_4)_3 = 399,91$.

Problem 2. A cobalt mixture ²

A freshly prepared mixture of Co, CoO and Co_2O_3 was heated under hydrogen atmosphere. A mass of 5,718 g of the mixture produced 1,221 g of water. When the same mass was reacted with excess of copper(II) sulfate solution, 5,794 g of a solid mixture were obtained. Determine the mass percentage of each component in the mixture.

Problem 3. Graphite Compounds I ^{3*}

The large distance between the carbon atoms layers in the graphite is due to the bonding between the layers is relatively weak. Because of this property, a variety of substances are able to intercalate within the layers originating laminar compounds of variable composition. The first intercalation compound with alkaline metals was reported in 1926: KC_8 , a bronze-colored solid that was prepared by the direct reaction between graphite and potassium vapors at 300 °C. This compound reacts with transition metal salts in tetrahydrofuran (solvent) at room temperature, according to the following reaction:



¹ Cuban National Chemistry Contest 2017. Day 1. Common exam. Problem 3.

² Cuban National Chemistry Contest 2009. Day 1. Common exam. Problem 2.

³ Cuban National Chemistry Contest 2014. Day 2. Individual exam. 10th and 12th grades. Problem 1.

When KC_8 is reacted with the metal chlorides **A**, **B**, **C** and **D**, compounds **E**, **F**, **G** and **H** are formed, respectively. It is known that the content of hydration water is different for each chloride and that the title metals belong to the same transition series.

Compounds	A	E	B	F	C	G	D	H
% of metal (in mass)	34,43	16,23	37,27	24,85	27,76	22,23	24,77	23,47

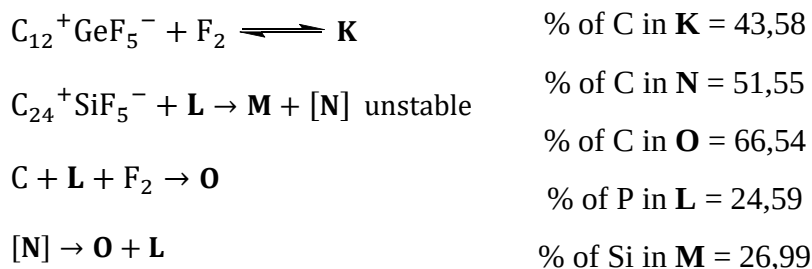
3.1. Calculate the empirical formula of **A**, **B**, **C**, **D**, **E**, **F**, **G** and **H**.

For the reaction to proceed with non-oxidant chlorides, the presence of chlorine is required. In this case, a complete charge transfer from the graphite to the hosted species occurs, with the formation of ions. Intercalation of graphite with FeCl_3 and AlCl_3 in the presence of chlorine originated compounds **I** and **J**, respectively. In both substances, there is a “cation” composed exclusively by carbon atoms. In **I** one electron is transferred per every 70 carbon atoms, while in **J** one electron is transferred per every 27 carbon atoms.

3.2. Calculate the formulas of **I** and **J** and the oxidation state of all the atoms forming these compounds. **I**: % of C 46,35 $n(\text{C}) : n(\text{Fe}) = 35:3$; **J**: % of C 42,68 $n(\text{C}) : n(\text{Al}) = 9:1$

Problem 4. Graphite compounds II ^{4*}

The large distance between the carbon atoms layers in the graphite is due to the bonding between the layers is relatively weak. Because of this property, a variety of substances are able to intercalate within the layers originating laminar compounds of variable composition. These compounds participate in several interchange and oxidation-reduction reactions. Some reactions of such compounds are summarized below:



⁴ Cuban National Chemistry Contest 2014. Reserved problem.

Compounds **L** and **M** are constituted by only two chemical elements. All compounds (**K**, **L**, **M**, **N** and **O**) contain fluorine.

4.1. Calculate the chemical formulas of compounds **K**, **L**, **M**, **N** and **O**. State the oxidation number of the atoms.

4.2. Balance the previous chemical equations.

Oxoacids intercalation under oxidizing conditions creates the so-called graphite salts, some of which are known from a century ago and have been very well characterized to date. Graphite oxidation at air in the presence of concentrated sulfuric acid forms the blue compound **P**. Similarly, the reaction between graphite and nitric acid forms **Q**, also colored blue and with a similar composition. Tanto **P** and **Q** contain the same “cation”, exclusively formed by carbon atoms.

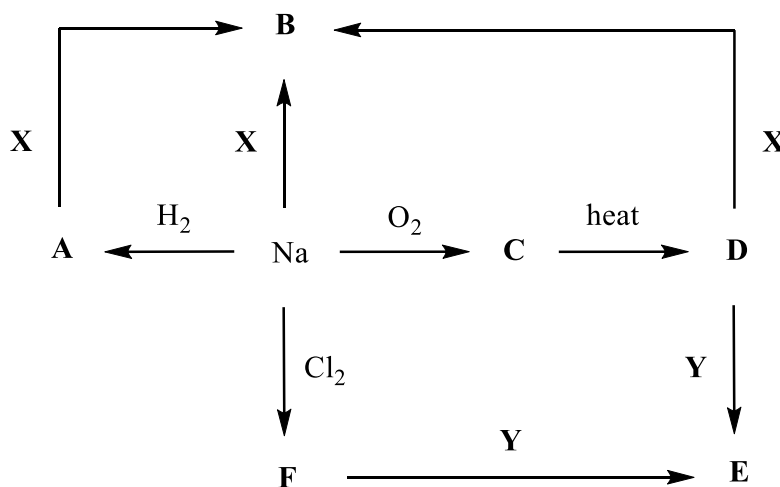
4.3. Find the empirical formula of **P** and write the balanced chemical equation for its obtention.

P: % of C 49,57 % of S 16,54 % of O 33,02

4.4. Find the empirical formula of **Q**. **Q:** % of C 60,52

Problem 5. Sodium compounds ^{5*}

All substances from **A** to **F** in the following scheme contain sodium.



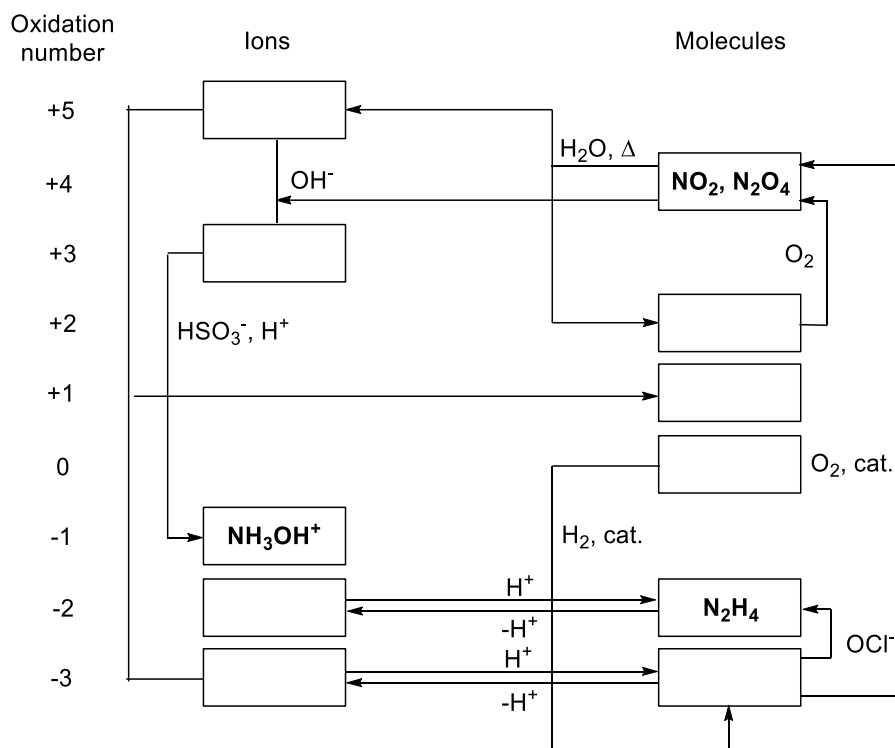
5.1. Write the formulas of substances **A**, **B**, **C**, **D**, **E**, **F**, **X** and **Y**.

5.2. Write the equations corresponding to all the chemical reactions represented in the scheme.

⁵ Cuban National Chemistry Contest 2017. Day 2. Individual exam. 10th grade. Problem 5.

Problem 6. Nitrogen compounds ^{6*}

The following scheme summarizes the interconversion processes between the main nitrogen-containing molecular and ionic species.



- 6.1. Complete the scheme with the formulas of the missing species.
- 6.2. Write balanced equations for all chemical transformations illustrated in the diagram.
- 6.3. Draw Lewis formulas for each nitrogen-containing species and the corresponding resonance structures, if any.
- 6.4. Considering the postulates of the VSEPR model, predict the geometry and the bond angle of the chemical species.

⁶ Cuban National Chemistry Contest 2014. Day 1. Common exam. Problem 2.

Problem 7. Chemical bonding ^{7*}

Non-metallic chemical element X forms oxides **A** and **B**. The ratio between the mass percentage of X in **A** and the mass percentage of X in **B** equals to 1,25. When the previous oxides react with water oxoacids **C** (39,1% in mass of X) and **D** (32,7% in mass of X) are formed, respectively, being **D** the most acidic of them.

- 7.1. Find which is the element X. Justify numerically.
- 7.2. Write the formula of compounds **A**, **B**, **C** and **D**.
- 7.3. The high acidity of **D** can be explained due to the instability of the mononegative anion formed when losing one H^+ ion. Represent the resonant structure of the title anion.
- 7.4. Predict the existence and the magnetic properties of the molecule X_2 . Use the postulates of the Molecular Orbital Theory.
- 7.5. Represent the electronic structure of **A** molecule according to the Valence Bond Theory employing hybrid orbitals.
- 7.6. In addition, element X forms ions $\text{X}_2\text{O}_3^{2-}$ and $\text{X}_2\text{O}_8^{2-}$. Represent Lewis structures for both ionic species.
- 7.7. When substance XCl_4 reacts with oxygen **E** is obtained, with formula XOCl_2 . Draw the Lewis structure of **E**. Predict the molecular geometry and the bond angle.

Problem 8. Identification of inorganic substances I ⁸

Six diluted solutions of the following substances were given: chromium(III) chloride, iron(III) chloride, hydrogen peroxide, sodium hydroxide and ammonia, in flasks marked with letters **A-F**. Reactions were carried out between all solutions. The results can be found below. Only the cases when there was a chemical reaction are given.

D with **A**: Formation of a greenish-grey precipitate which is soluble in excess of **D**.

E with **A**: Formation greenish-grey precipitate which is soluble in excess of **E**.

⁷ Cuban National Chemistry Contest 2017. Day 2. Individual exam. 10th grade. Problem 7.

⁸ National German Competition 2012. Problem 3-17.

D with **B**: Formation of a green precipitate.

E with **B**: Formation of a green precipitate which is soluble in excess of **E**.

D with **C**: Formation of a gas.

E with **C**: Formation of a gas.

F with **D**: Formation of a brown precipitate.

F with **E**: Formation of a brown precipitate.

8.1. Assign to each letter the corresponding substance.

8.2. Write the equations corresponding to all chemical reactions that happen.

8.3. Predict the color of the starting solutions.

Problem 9. Identification of inorganic substances II ⁹

There are three mixtures containing pulverized inorganic solids. These mixtures may contain the following substances.

Mixture A: $(\text{NH}_4)_2\text{CO}_3$, AgNO_3 , $\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$, NH_4NO_3 , $\text{NiCl}_2 \cdot 2\text{H}_2\text{O}$

Mixture B: ZnO , KI , $\text{Pb}(\text{NO}_3)_2$, BaSO_4 , MnO_2 , Mg

Mixture C: CaCO_3 , NH_4I , $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$, SiO_2 , $\text{CuCl}_2 \cdot 2\text{H}_2\text{O}$.

9.1. Determine which compounds appear in each mixture and which do not, given the following experimental results.

Mixture A: It is colorless. When treated with HCl or with HNO_3 no gas evolution is observed. Only a fraction is soluble in water, and the white residue is soluble in the presence of NH_3 .

Mixture B: It is grey. When adding water, there is a grey residue that does not dissolve. The mixture dissolves slowly in HNO_3 without leaving a residue and with gas evolution. When HCl is added, a white precipitate is observed.

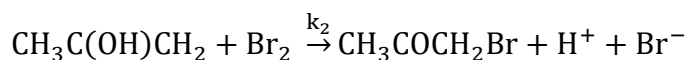
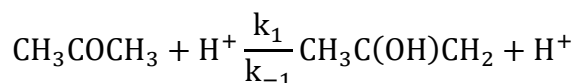
Mixture C: It is green colored. It dissolves both in HCl and in HNO_3 with gas evolution. The resultant solution is blue. When treated with ammonia, an intense blue is obtained.

⁹ International Chemistry Olympiad 2008. Preparatory Problems. Problem 31.

9.2. Write the equations corresponding to all the chemical reactions that happen.

Problem 10. Analysis of reaction mechanisms *

The following mechanism corresponds to the bromination of acetone.

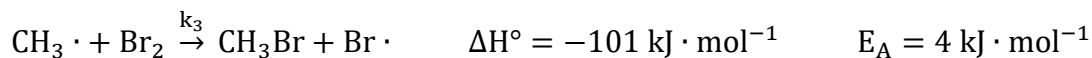
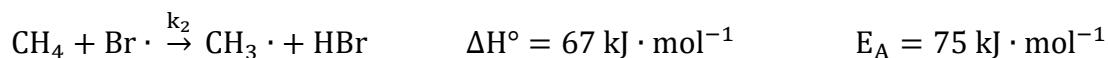
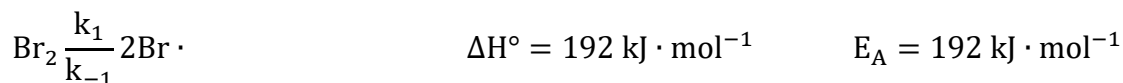


10.1. Write the total equation of the reaction.

10.2. Find an expression for the rate of formation of the main product of the reaction applying the steady state approximation.

10.3. ¿Under what experimental conditions a pseudo-order behavior can be observed?

The following mechanism corresponds to the reaction of bromination of methane.



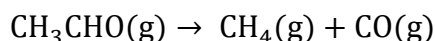
10.4. Find the total equation of the reaction.

10.5. Find the rate limiting step and its molecularity.

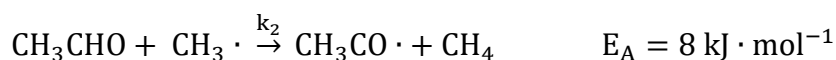
10.6. Draw an approximate diagram of energy vs reaction coordinate. In the scheme, mark reactants, intermediates, products, activation energy and heat of reaction.

Problem 11. Ethanal decomposition *

In air absence, pyrolysis or thermic decomposition of ethanal produces methane and carbon monoxide. During the reaction ethane is also formed.



A simplified mechanism was proposed to explained these experimental results by Rice-Herzfeld, shown below.



11.1. Find an expression for the rate of methane formation.

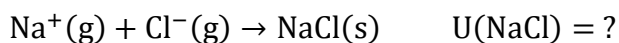
11.2. Relate the overall rate constant with the rate constants of the individual reaction steps of the mechanism.

11.3. Relate the overall activation energy with the activation energies of the individual reaction steps of the mechanism.

Problem 12. Born-Haber cycle ^{10*}

Thermodynamic cycles allow to calculate enthalpy changes associated to process that are difficult to implement in practice. This is possible because the enthalpy is a state function and only depends on the initial and final states.

12.1. Use the given data to calculate the lattice enthalpy U of NaCl.



Standard formation enthalpy:



¹⁰ Cuban National Chemistry Contest 2017. Day 2. 10th grade. Problem 6. (Fragment)

Standard dissociation enthalpy:



Electron affinity:



Standard sublimation enthalpy:



Ionization energy:

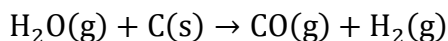


12.2. Substances AgCl and NaCl present the same crystalline structure in the solid state. Both are constituted by the same anion and have cations with equal charges and similar sizes ($r(\text{Ag}^+) = 115 \text{ pm}$, $r(\text{Na}^+) = 102 \text{ pm}$). How do you explain that the lattice enthalpy of both compounds are so different? $U(\text{AgCl}) = -912 \text{ kJ} \cdot \text{mol}^{-1}$.

12.3. Compare the solubility in water and the melting point of substances NaCl y AgCl.

Problem 13. Coal gasification ¹¹

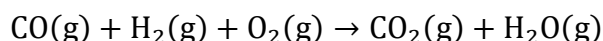
In the process of coal gasification, coal is converted in coal gas, a flammable fuel containing carbon monoxide and hydrogen



13.1. Calculate the standard enthalpy change for the reaction from the following chemical equations and the standard enthalpy changes.



Coal gas can be used as fuel

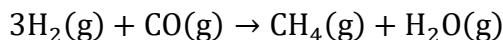


13.2. Calculate the standard enthalpy change for the combustion of coal gas using the additional information given

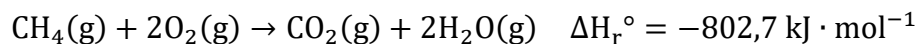


¹¹ International Chemistry Olympiad. Preparatory Problems. Problem 4.

Coal gas can also carry out the methanation process

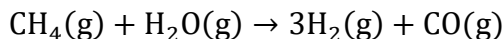


13.3. Calculate the standard enthalpy change for the methanation of coal gas, with the additional information given



Problem 14. Industrial preparation of hydrogen ¹²

Hydrogen can be industrially prepared by heating hydrocarbons such as methane with water at atmospheric pressure:



14.1. Given the following thermodynamic data, calculate $\Delta G_{\text{r}}^{\circ}$ for the previous reaction at 298 K and the value of the equilibrium constant K_{p} .

	$\text{CH}_4(\text{g})$	$\text{H}_2\text{O}(\text{g})$	$\text{H}_2(\text{g})$	$\text{CO}(\text{g})$
$\Delta H_{\text{f}}^{\circ}(298 \text{ K}) / \text{kJ} \cdot \text{mol}^{-1}$	-74,4	-241,8		-110,5
$S^{\circ}(298 \text{ K}) / \text{J} \cdot \text{K}^{-1} \cdot \text{mol}^{-1}$	186,3	188,8	130,7	197,7

14.2. How the equilibrium constant changes with temperature?

The industrial preparation can be carried out at atmospheric pressure and high temperatures, without catalyst. Typically, 0,2% volume of methane gas remains in the equilibrium mixture.

14.3. Assuming that the reaction was started with equal volumes of methane and water vapor, calculate the value of K_{p} for the industrial process giving 0,2% volume of methane gas in the equilibrium.

14.4. Use the previous results and the integrated form of the van't Hoff's isochore to estimate the temperature employed in the industry for the preparation of hydrogen from methane.

¹² International Chemistry Olympiad 2009. Preparatory Problems. Problem 5.

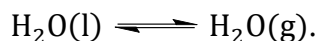
Problem 15. Hydrates ¹³

If you prepare blue crystals of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ to form an aqueous solution of copper(II) sulfate and store them in an open container inside a hot and dry room, the edges will become light blue or white colored, they will “weather” slowly. If you want to keep the blue color of the crystals, you must cover them with a transparent lacquer. There are several copper(II) sulfate hydrates $\text{CuSO}_4 \cdot n\text{H}_2\text{O}$ ($n = 1, 3, 5$), which interconvert depending on the content of water vapor in the air.

The amount of water in the air is usually reported as relative humidity H_r :

$$H_r(\text{in } \%) = \frac{p(\text{H}_2\text{O})}{p^\circ(\text{H}_2\text{O})} \cdot 100$$

Where $p(\text{H}_2\text{O})$ is the vapor pressure of water and $p^\circ(\text{H}_2\text{O})$ is the pressure when the air is saturated with water vapor (when the relative humidity of the air is 100%) at the same temperature. $p^\circ(\text{H}_2\text{O})$ is, by definition, the pressure of water vapor when the following equilibrium is established



15.1. Calculate the water pressure $p^\circ(\text{H}_2\text{O})$ at 25 °C.

15.2. Calculate the value of the relative humidity of the air (in %) at which copper(II) sulfate hydrates with $n = 3$ and $n = 5$ interconvert at 25 °C.

15.3. In which form will you find a crystal of copper(II) sulfate in Cuba in winter? Typical weather conditions in this season after the passage of a cold front are 25 °C temperature and 50% relative humidity.

15.4. Calculate the boiling point of water at the atmospheric pressure. Consider that the atmospheric pressure is equal to the standard pressure ($p = p^\circ = 100 \text{ kPa}$).

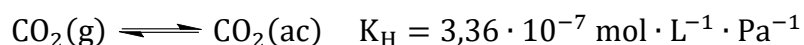
Compound	$\Delta H_f^\circ(298 \text{ K})/\text{kJ} \cdot \text{mol}^{-1}$	$S^\circ(298 \text{ K})/\text{J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$
$\text{CuSO}_4 \cdot 5\text{H}_2\text{O} (\text{s})$	-2278,0	305,4
$\text{CuSO}_4 \cdot 3\text{H}_2\text{O} (\text{s})$	-1683,1	225,1
$\text{CuSO}_4 (\text{s})$	-771,4	109,2
$\text{H}_2\text{O} (\text{l})$	-285,8	70,1
$\text{H}_2\text{O} (\text{g})$	-241,8	188,7

¹³ Cuban National Chemistry CONtest 2017. Day 2. 11th grade. Problem 8.

Problem 16. Acid-base equilibrium ^{14*}

In contrast with pure water, which is neutral, rainwater is slightly acidic. Some causes for this phenomenon are natural and others are triggered by human action. Natural rainwater contains dissolved carbon dioxide in the air.

16.1. Calculate the pH of natural rainwater. The air contains about 0,035% (v/v) of carbon dioxide and the average atmospheric pressure is 101,3 kPa. $K_{a1}(\text{CO}_2) = 4,47 \cdot 10^{-7}$; $K_{a2}(\text{CO}_2) = 4,68 \cdot 10^{-11}$



In the air, sulfur dioxide and nitrogen monoxide waste produced by industrial activity are oxidized to sulfur trioxide and nitrogen dioxide, respectively. These gasses react with water to form sulfuric and nitric acids. These reactions provoke that the so-called acid rain has an average pH value of about 4,5.

16.2. Write the chemical equation for the reaction between nitrogen dioxide and water.

16.3. Calculate which concentration of nitric acid causes a pH of 4,5.

16.4. Calculate which concentration of sulfuric acid ($K_{a2} = 1,02 \cdot 10^{-2}$) causes a pH of 4,5.

Some caves are formed when the water filtered through the fractures (which is enriched in CO_2 content) slowly dissolves the rock in a process that can last for millions of years. This process also creates rock formations such as stalactites and stalagmites. These caves are formed in calcareous massifs of limestone and dolomites. The main components of the rocks are calcite (CaCO_3) and dolomite ($\text{CaMg}(\text{CO}_3)_2$), respectively. The most extensive cave system of this kind is Mammoth Cave, in the United States, with more than 590 km of interconnected galleries. Santo Tomás cave system, in Sierra de los Órganos, in Pinar del Río province, is the largest from Cuba, with an extension of 46,2 km and is also formed by such caves. Its exploration started in 1954 by the Cuban Speleological Society under the direction of the eminent Cuban scientist, patriot and politician Antonio Núñez Jiménez (1923-1998).

16.5. Write the chemical equation for the reaction between carbon dioxide dissolved in natural rainwater and dolomite.

¹⁴ Cuban National Chemistry Contest 2017. Day 2. 11th grade. Problem 9.

16.6. Calculate the solubility (in $\text{mol}\cdot\text{L}^{-1}$) of calcite in natural rainwater. Assume that the natural rainwater has a pH equal to 5,65. $K_{\text{ps}}(\text{CaCO}_3) = 3,36\cdot 10^{-9}$.

Problem 17. Acid-base titrations^{15*}

Acid-base titrations are methods widely used in the chemical laboratory. They are based in the acid-base or neutralization reaction between the analyte (the substance whose concentration is unknown) and the standard (the substance whose concentration is known by the analyst). In the following two examples these methods are applied.

A- Calcium oxide was added to pure water and the resultant suspension was let to stand for a week in a closed plastic container, under inert atmosphere. The suspension was filtered to eliminate the formed solid and then stored for its posterior analysis. Immediately, several 20,00 mL aliquots from the filtered solution were titrated with hydrochloric acid previously standardized to a concentration of $0,0150 \text{ mol}\cdot\text{L}^{-1}$ in the presence of phenolphthalein as indicator. The volume consumed from the titrant were 30,5; 29,4; 29,8 and 29,6 mL.

17.1. In the previously described procedure several measurements were taken in order to avoid the introduction of errors. Why is it necessary to store the problem solution in a plastic container and under inert atmosphere? Why was the suspension allowed to stand for a week?

17.2. Describe the working principle of an acid-base indicator.

17.3. Before carrying out the titration, it is necessary to standardize the hydrochloric acid due to this substance is not a primary standard. Which characteristics possess such substances?

17.4. Calculate the solubility product constant of calcium hydroxide.

B- A student took an aliquot of 5,00 mL from the electrolytic solution of a lead battery and diluted it with water until completing 2,00 L. Next, the student took 10,00 mL from the diluted solution, pour them in a beaker with a mechanical stirrer and titrated with $\text{NaOH } 0,01000 \text{ mol}\cdot\text{L}^{-1}$. The student made a titration curve with the aid of a pH-meter and though the graphical method, the student determined that the volume of titrant consumed up to the equivalence point was 24,14 mL.

¹⁵ Cuban National Chemistry Contest 2017. Day 2. 12th grade. Problem 13. (Fragment and adaptation)

17.5. Calculate the molar concentration of the sulfuric acid inside the battery. Ignore the influence of the dissolved lead.

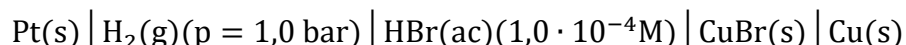
17.6. Sketch the approximate titration curve that was obtained by the student. For this purpose, calculate the pH in the titration flask after the addition of: 0, 12,07, 24,14 and 36,21 mL of NaOH. $K_{a2}(\text{H}_2\text{SO}_4) = 1,02 \cdot 10^{-2}$, $K_W = 1,00 \cdot 10^{-14}$.

17.7. Suppose that the student wants to visually detect the end point of the titration. Select the **most appropriate** acid-base indicator that should be use in such case.

Indicator	Methyl orange	Methyl red	Blue bromothymol	Phenolphthalein	Thymolphthalein
Color change interval	3,2-4,4	4,8-6,0	6,0-7,6	8,2-10,0	9,4-10,6

Problem 18. Solubility of CuBr ¹⁶

The electromotive force of the following cell is 0,559 V at 298 K. Assume that all the species within the cell behave ideally.



18.1. Write the cathodic and anodic half equations, the total equation of the cell and the corresponding Nernst expression.

18.2. Calculate the standard potential for the CuBr electrode.

The standard electrode potential for the $\text{Cu}^+(\text{ac})/\text{Cu(s)}$ pair is 0,522 V.

18.3. Calculate ΔG° for the dissolution of CuBr at 298 K and the solubility product constant of CuBr.

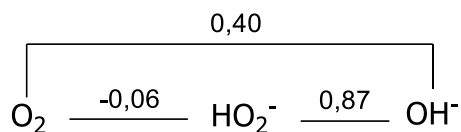
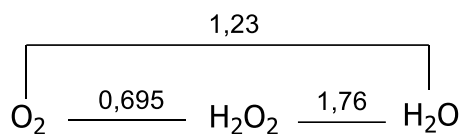
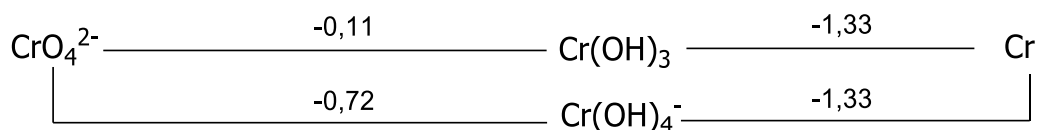
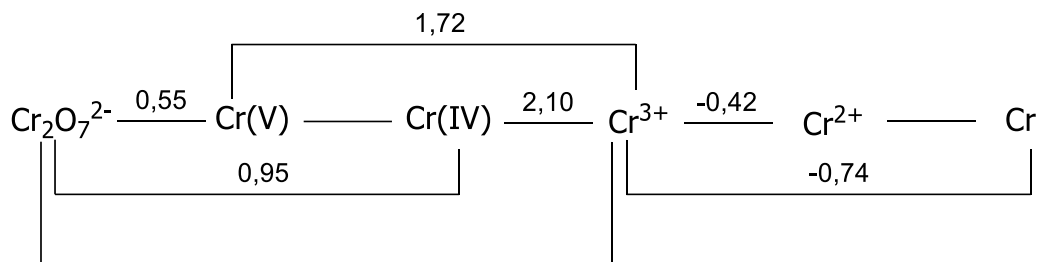
18.4. Calculate the concentration of $\text{Cu}^+(\text{ac})$ ions present in the cell.

18.5. How much will the EMF of the cell will change if the hydrogen pressure is doubled?

¹⁶ International Chemistry Olympiad 2009. Preparatory Problems. Problem 13.

Problem 19. Electrode potential charts ¹⁷

The Latimer diagrams for the chromium and oxygen species in acid media (pH = 0) and basic media (pH = 14) are shown below.



19.1. Calculate the values missing in the diagram.

19.2. Which chromium species are unstable and disproportionate? Calculate the equilibrium constant for the disproportionation of Cr^{2+} ion.

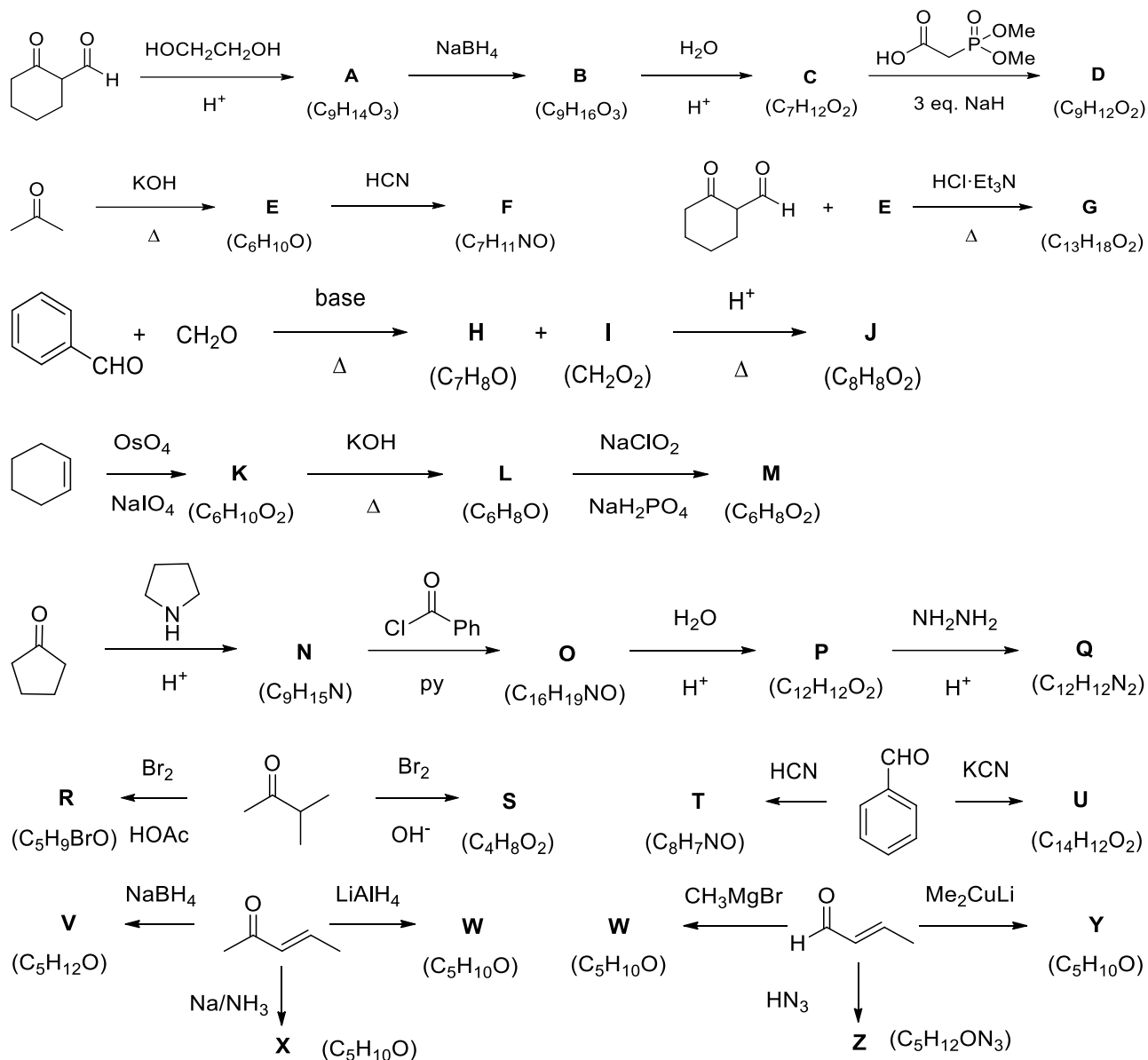
19.3. Calculate the solubility product constant of chromium(III) hydroxide.

19.4. What would happen if the pH of the solution containing chromate(VI), Cr(III) and hydrogen peroxide to zero? What would happen if the pH is adjusted to 14? Write the equations of the reactions that happen and calculate the corresponding standard potential changes.

¹⁷ International Chemistry Olympiad 2008. Preparatory Problems. Problem 3. (Fragment)

Problem 20. Aldehydes and ketones *

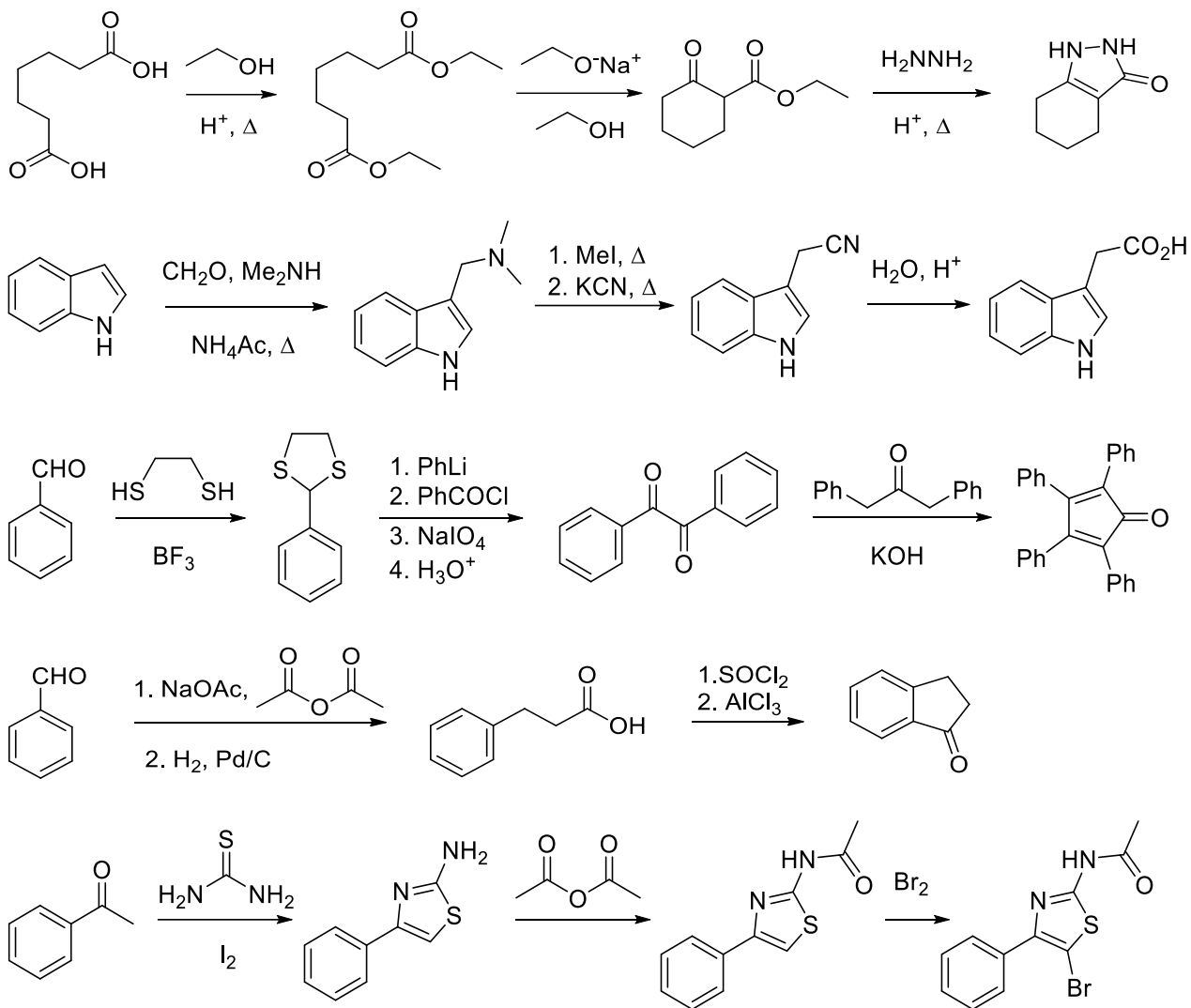
Aldehydes and ketones present a high importance in organic synthesis. These compounds take part in four fundamental kinds of chemical reactions, namely: oxidation-reduction, nucleophilic addition, nucleophilic substitution and reactions through the α carbon. Several examples of these kind of reactions are given below.



19.20 Represent the structures of compounds A-Z.

Problem 21. Reaction mechanism *

21.1. Suggest mechanisms for all the reactions in the following synthetic sequences.



Annex 1. Results of the Cuban National Chemistry Contest 2017 ¹⁸

Results by provinces

Province	G	S	B	PSN	Points	Place
LT	4	6	0	9	56	1
MT	5	1	2	6	42	2
CM	1	7	2	4	36	3
SS	2	2	4	4	28	4
LH	1	2	7	2	22	5
PR	0	4	3	1	17	6
SC	1	2	2	1	15	7
GR	1	0	3	1	10	8
VC	0	1	6	0	9	9
GT	0	0	5	0	5	10
AR	0	0	2	0	2	11
CF	0	0	1	0	1	12
MY	0	0	0	0	0	13
CA	0	0	0	0	0	13
HO	0	0	0	0	0	13
IJ	0	0	0	0	0	13

Qualification tribunal

MSc. Rolando Alfonso Valdés	VC	IPVCE Ernesto Che Guevara
MSc. Julio Rodríguez Benítez	CM	IPVCE Máximo Gómez Baéz
MSc. Agustín Plasencia Calero	SS	IPVCE Eusebio Olivera
MSc. Juan Francisco Olivera	PR	IPVCE Federico Engels
MSc. Asterio Marrero Machado	GR	IPVCE Silberto Álvarez Aroche
MSc. Rafael León Pérez	LH	IPVCE Vladimir I. Lenin
MSc. Luis Enrique Guerra Castaño	LT	IPVCE Vladimir I. Lenin
Lic. Jorge Alberto Arias Rodríguez	SC	IPVCE Antonio Maceo Grajales
MSc. José María Fernández García	AT	Universidad Tecnológica de La Habana
MSc. Tomas Sánchez Herrera	GT	IPVCE José Maceo Grajales

Authors of the problems

MSc. Gerardo M. Ojeda Carralero
BSc. Leonardo González Ceballos

Revision

BSc. Asiel Mena Jiménez
Javier E. Alfonso Ramos

¹⁸ Symbology:

G: Gold Medal (5 points)

S: Silver Medal (3 points)

B: Bronze Medal (1 point)

PSN: National Pre-selection (2 puntos)

Tenth grade

Rey Michel Rodríguez Navarro	CM	IPVCE Máximo Gómez Báez	96,00	G
Davier Sánchez Bello	MT	IPVCE Carlos Marx	80,00	G
Eduardo Addiel Landrove Escalona	LT	IPVCE Luis Urquiza Jorge	73,00	G
Massiel B Fuentes Vázquez	MT	IPVCE Carlos Marx	72,50	G
Osniel A García Orihuela	MT	IPVCE Carlos Marx	71,10	G
Rachel Yina Báez González	LT	IPVCE Luis Urquiza Jorge	71,00	G
Maday Sánchez Orama	MT	IPVCE Carlos Marx	71,00	G
Jordan Lismar López Mantecón	CM	IPVCE Máximo Gómez Báez	69,00	S
Marco Antonio Fernández Roquero	CM	IPVCE Máximo Gómez Báez	65,00	S
José Manuel Cabrera Pereira	SS	IPVCE Eusebio Olivera Rodríguez	64,50	S
Ana Elia Miranda Castañeda	PR	IPVCE Federico Engels	62,10	S
Lucía María Figueroa Torres	CM	IPVCE Máximo Gómez Báez	57,10	S
Liliana Llamas Pérez	SC	IPU Rafael María de Mendive	54,00	S
José Angel Delgado Morejón	LH	IPVCE Vladimir I. Lenin	53,00	S
Lisvany Martín Rodríguez	VC	IPVCE Ernesto Guevara	51,50	S
Lester Padron Cuella	PR	IPVCE Federico Engels	51,10	S
René Alejandro Vuelta Garcés	SC	IPU Rafael María de Mendive	51,00	S
Laura Barreto Gonzalez	AR	IPVCE Mártires de Humboldt 7	50,50	B
Yesenia de la C. Valladares Suárez	PR	IPVCE Federico Engels	50,00	B
Beatriz Reyes Manals	SC	IPVCE Antonio Maceo Grajales	50,00	B
Anier Alcides Peña Omar	GT	IPVCE José Maceo Grajales	49,10	B
Javier Alejandro Sánchez Sayas	SS	IPVCE Eusebio Olivera Rodríguez	48,50	B
Luis Fernando García Aguilar	PR	IPVCE Federico Engels	47,10	B
Cecilia Ribeaux Villalón	GT	IPVCE José Maceo Grajales	47,00	B
Rosa María Mederos Martínez	MT	IPU Martin Klein	47,00	B
Roxana Penha Mendieta	LH	IPVCE Vladimir I. Lenin	45,40	B
Rafael Alejandro Perurena García	SS	IPVCE Eusebio Olivera Rodríguez	45,10	B
Camilo Mas Jacobs	LH	IPVCE Vladimir I. Lenin	45,00	B
Erik Richard Ventura Castro	SS	IPVCE Eusebio Olivera Rodríguez	43,10	B
David Linares Ramírez	VC	IPU Capitán Roberto Rodríguez	42,50	B
Zujay Bell Larduet	SC	IPU Cuqui Bosh	41,10	B
Dairon A. Pérez Sánchez	VC	IPVCE Ernesto Guevara	41,10	B
Amanda I Rivero Arbolay	VC	IPVCE Ernesto Guevara	40,60	B
Geidi Roxana Vargas Herryman	GR	IPVCE Silberto Álvarez Aroche	40,00	B
Laura Gil Espenisa	LH	IPVCE Vladimir I. Lenin	36,50	B

Eleventh grade

Isaac Enrique Peñuela Puente	SC	IPVCE Antonio Maceo Grajales	85,50	G
Carlos Rafael Castillo Hernández	SS	IPVCE Eusebio Olivera Rodríguez	84,50	G
Lilisbeth de la C. Bénitez Rojas	LT	IPVCE Luis Urquiza Jorge	81,00	G
Sergio Modesto Reyes Díaz	LH	IPVCE Vladimir I. Lenin	80,00	G
Adolfo Alejandro Cruz Cruz	SS	IPVCE Eusebio Olivera Rodríguez	78,00	G
Sara María Collejo Fernández	LT	IPVCE Luis Urquiza Jorge	73,50	S
Josué Ponce de León Cabrera	CM	IPVCE Máximo Gómez Báez	71,50	S
Adiel Alexis Martínez Rodríguez	MT	IPVCE Carlos Marx	71,00	S
Ariel Ernesto Martínez Suffos	CM	IPVCE Máximo Gómez Báez	70,50	S
Maricary Sifontes Niebla	SS	IPVCE Eusebio Olivera Rodríguez	70,50	S
Lilianne Gil Batista	LT	IPVCE Luis Urquiza Jorge	70,00	S
Denys Daniel Núñez Pupo	LT	IPVCE Luis Urquiza Jorge	68,00	S
Ariel Alejandro Castro Sifonte	CM	IPVCE Máximo Gómez Báez	67,50	S
Gricela María Águila Álvarez	PR	IPVCE Federico Engels	61,00	S
Pablo Enrique Remedios Rosa	SS	IPVCE Eusebio Olivera Rodríguez	54,50	B
Yainel Miranda Hernández	PR	IPVCE Federico Engels	53,00	B
Manuel A. Santana Hernández	VC	IPVCE Ernesto Guevara	52,50	B
Liliana Matos Ávila	LH	IPVCE Vladimir I. Lenin	50,50	B
Jaime Reina Alares	LH	IPVCE Vladimir I. Lenin	47,50	B
Rolando Pérez Heredia	GT	IPVCE José Maceo Grajales	44,50	B
Noel G. Niebla Gómez	CF	IPVCE Carlos Roloff	38,50	B
Jorge Ernesto Claro Díaz	GT	IPVCE José Maceo Grajales	38,00	B
Claudia Natalí Jeréz Zamora	GR	IPVCE Silberto Álvarez Aroche	34,50	B
Marcos A González Correa	MT	IPVCE Carlos Marx	33,00	B
Elizabeth Teresita Cruz Portal	CM	IPVCE Máximo Gómez Báez	32,50	B
Karen Valdés Delgado	AR	IPVCE Mártires de Humboldt 7	32,00	B

Twelfth grade

Leonor Enedis Collejo Fernández	LT	IPVCE "Luis Urquiza Jorge"	87,00	G
José Rafael Rodríguez Rodríguez	MT	IPVCE "Carlos Marx"	86,60	G
Norberto Javier Tamayo León	GR	IPVCE "Silberto Álvarez Aroche"	86,00	G
Eduardo Eloy Blanco Rodríguez	PR	IPVCE "Federico Engels"	80,90	S
Patricia Cartaya Quintero	LT	IPVCE "Luis Urquiza Jorge"	77,30	S
Nataly Rodríguez González	LT	IPVCE "Luis Urquiza Jorge"	76,60	S
Vanessa Mabel Martínez Núñez	LT	IPVCE "Luis Urquiza Jorge"	76,50	S
Cesar Eduardo Escalante Bermúdez	LH	IPVCE "Vladimir I. Lenin"	74,10	S
Lídice Cabrera Victoria	CM	IPVCE "Máximo Gómez Báez"	71,40	S
Adonis Castillo Ledesma	LH	IPVCE Vladimir I. Lenin	70,10	B
Leonardo Rafael Rodríguez González	CM	IPVCE Máximo Gómez Báez	68,70	B
Rolando Estévez Mendoza	GT	IPVCE José Maceo Grajales	67,70	B
Jorge Ernesto Mato Frontela	LH	IPVCE Vladimir I. Lenin	63,60	B
Stany Almaguer Arias	GR	IPVCE Silberto Álvarez Aroche	58,80	B
Michel A. Machado León	VC	IPVCE Ernesto Guevara	55,90	B
Melissa Gómez Saucedo	VC	IPVCE Ernesto Guevara	53,40	B

Annex 2. Results in the international events in 2016

XXI Iberoamerican Chemistry Olympiad

Celebrated on September 19 to 24, 2016 at the National University of Colombia, in Bogota. 18 countries from Latin America and the Iberic Peninsula took part in the competition. The members of the Cuban delegation were:

MSc. Julio Rodríguez Benítez (Mentor) IPVCE "Máximo Gómez Báez", Camagüey	
Miguel Alejandro González Martínez (Student) IPVCE "Eusebio Olivera Rodríguez", Sancti Spíritus	Silver Medal
Liván Borrego Camejo (Student) IPVCE "Mártires de Humboldt 7", Artemisa	Bronze Medal
Fernando Gabriel Martín Rodríguez (Student) IPVCE "Máximo Gómez Báez", Camagüey	Bronze Medal
Norberto Javier Tamayo León (Student) IPVCE "Silberto Álvarez Aroche", Granma	Honorary Mention



Cuban team taking part in the Iberoamerican Chemistry Olympiad (from left to right):

Norberto, Fernando, Julio, Liván y Miguel.

X Central-American and VIII Caribbean Chemistry Olympiad

Celebrated on July 18 to 22, 2016 at the University of Havana, in Cuba. Five countries took part in the event. The members of the Cuban delegations were:

MSc. Rolando Alfonso Valdés (Mentor)
IPVCE "Ernesto Guevara", Villa Clara

MSc. Agustín Plascencia Calero (Mentor)
IPVCE "Eusebio Olivera Rodríguez", Sancti Spíritus

José Rafael Rodríguez Rodríguez (Student) Gold Medal
IPVCE "Carlos Marx", Matanzas

Kendria Beatriz Góngora Parra (Student) Gold Medal
IPVCE "Luis Urquiza Jorge", Las Tunas

Leonor Enedis Collejo Fernández (Student) Silver Medal
IPVCE "Luis Urquiza Jorge", Las Tunas

Eliosdanis González Machado (Student) Silver Medal
IPVCE "Silberto Álvarez Aroche", Granma



Cuban students Estudiantes cubanos (left to right):
Rafael, Kendria, Leonor, Eliosdanis.

Cuban members of the Scientific and Organizer
Committee, guides and mentors (above); and instructors
of the practical exam (below).

Annex 3. Syllabus of the National Chemistry Contest for High School

Tenth grade

Topic I. Substances and chemical reactions

1. Nomenclature and notation of quaternary compounds.
2. Adjusting chemical equations by the variation of oxidation numbers.
3. Law of conservation of mass, law of definite proportions and law of multiple proportions.
4. Chemical calculations related to stoichiometry and the concept of limiting substance.
5. Chemical calculations of samples.
6. Calculations to find empirical and molecular formulas.

Topic II. Hydrogen

1. Quantum numbers. Slater's rules. Relation between electronic structure and Periodic Table. Variation of atomic properties in the Periodic Table. Chemical periodicity (including diagonal periodicity).
2. Chemical bonding. Valence bond theory (Lewis's formulas). Valence Shell Electron Pair Repulsion (VSEPR) model. Molecular Orbital Theory (MOT) by Linear Combination of Atomic Orbitals (LCAO) method for diatomic molecules. Representation of hybridized molecules by the notation nl^x . Molecular geometry (including d orbitals).

Topic III. Thermic manifestations of chemical reactions

1. Zeroth Law of Thermodynamic. Temperature. State function.
2. Calculations related to the experimental determination of the reaction enthalpy using calorimeters.

Topic IV. Water

1. The same topic included in the subject received by the general course.

Topic V. The rate of chemical reactions

1. Order and molecularity. Methods to derive the reaction order. Experimental determination of the reaction order. Determination of the rate constants and the rate at specific concentrations of the reactants. Comparative analysis of simple mechanisms. Bodenstein's method to find the rate equation in complex reactions. Determination of the activation energy according to Arrhenius. Tema

Topic VI. Non-metals

1. Chemical periodicity. Application of the knowledge about chemical bonding and periodicity to non-metallic elements, their simple and compound substances.

Note: Content from ninth and eight grade are included.

Eleventh grade

Topic I. Solutions

1. Ideal solution. Solubility of gasses in liquids. Calculations.
2. Brief study of saline crystal hydrates. Solubility curves.
3. Aw of the chemical equivalents. Calculations related to (equivalent) molar concentrations and molality, also to mixture of solutions.
4. Colligative properties of non-electrolytes solutions. Calculations.
5. Colligative properties of electrolytes solutions. Van't Hoft's i factor. Calculations.

Topic II. Molecular equilibrium

1. Relation between K_c and K_p .
2. Brief thermodynamic study of a chemical system (ΔG , ΔS , ΔH). Relation between ΔG° and the equilibrium constant K_p .
3. Problems about equilibrium. Initial moles. Density of a mixture of gasses. More than one system in equilibrium. Application of L'Chatelier's principle. Use of the equation of Van't Hoff's isotherm. Graphical representation of experimental facts.

Note: Study indistinctly homogeneous and heterogeneous systems.

Topic III. Ionic equilibrium

1. Problems related to acids and bases that behave as weak electrolytes, pH and pOH.
2. Problems related to polyprotic acids.
3. Acid-base indicators. Calculations related to pK.
4. Buffer solutions. Calculations.
5. Saline hydrolysis. Calculations. (K_h , α_h , simple and double rupture hydrolysis).
6. K_{sp} . Problems.
7. Factors modifying the equilibrium state. Temperature (qualitative and quantitative analysis). Pressure (qualitative and quantitative analysis). pH (qualitative and quantitative analysis). Common ion effect (qualitative and quantitative analysis). Saline effect (qualitative analysis only).

8. Acid-base titrations: titration curves. Selection of the indicators for the titration.

Topic IV. Electrochemistry

1. Adjustment of redox equations.
2. Method of the variation of the degree of oxidation.
3. Ion-electron method.
4. Indeterminate coefficient method.
5. Nernst equation. Problems.
6. Problems related to the standard redox potential diagram of a species.
7. Concentration cells.
8. Electrolysis with saline solutions with inert and active electrodes.
9. Faraday's laws. Problems.

Topic V. Metals

1. Applications of the contents from the previous units to thesis once.

Note: Contents from topics I, II and V corresponding to tenth grade are included.

Twelve grade

Topic I. Group IV-A. Carbon and its compounds.

1. Relate the topic with the periodic table and stoichiometric chemical calculations.
2. Calculations to find empirical and molecular formula para determinar fórmula empírica y molecular.

Topic II. Saturated hydrocarbons

1. Conformational analysis (for example, butane).
2. Preparation methods. Reactions that maintain the same number of carbon atoms in the molecule: catalytic hydrogenation of alkenes and alkynes; from alkyl halides (hydrolysis of Grignard reagents, reduction of R-X with metal and acid, reduction of LiAlH_4); hydriodic reduction of alcohols, aldehydes, ketones and carboxylic acids; reduction of monocarboxylic acids with H_2 and Ni; Clemmensen reduction. Reactions that diminish the number of carbon atoms in the molecule: fusion of monocarboxylic acids salts with washing soda. Reactions that increase the number of carbon atoms in the molecule: Wurtz synthesis; reactions of Grignard reagents with RX; Kolbe's electrolytic method.

3. Chemical properties: halogenation. Free radical mechanism. Quantitative determination of the orientation in the halogenation. Stability of free radicals. Difference between reactivity and selectivity. Combustion. Pyrolysis.

Topic III. Stereoisomerism

1. Classification. Conformational isomers. Configurational isomers. Enantiomers. Asymmetric carbon and optical activity. Chirality. Chiral center. Flat formulas, Fischer's and Newman's projection, horse and wedge perspective representations. Enantiomers: physical and chemical properties. Diastereoisomers: physical and chemical properties. Racemate or racemic mixture. Specification of the configuration (relative D and L), (absolute R and S), priority order. Meso, eritro and threo structures.

Topic IV. Alkyl halides

1. Structure and nomenclature. Physical properties. Preparation of alkyl halides: halogenation of alkanes, alkanes and alkynes; reaction of alcohols with phosphorous halides and thionyl chloride; reaction of aldehydes and ketones with phosphorous pentachloride.

2. Chemical properties. Nucleophilic substitution reactions: reaction with KOH and NaOH in aqueous media; hydrolysis of R-X; reactions of alkoxides and phenoxides (Williamson's synthesis); metallic acetylides forming alkynes; metallic cyanides; NH_3 and amines (Hoffman's synthesis); argentate salts of carboxylic acids. Elimination reactions: dehydrohalogenation of monohalides and vicinal dihalides. Preparation of the Grignard's reagent. Wurtz's synthesis. Mechanism of nucleophilic substitution and elimination. The $\text{S}_{\text{N}}2$ reaction, stereochemistry and reactivity. $\text{S}_{\text{N}}1$ reaction, stereochemistry, reactivity and transposition. Difference among the different types of mechanism. E1 and E2 elimination, reactivity and evidence for both. Elimination vs substitution. Prediction of regioselectivity and stereoselectivity.

Topic V. Unsaturated hydrocarbon

1. Alkenes. Nomenclature. Geometrical isomerism. Physical properties.

2. Preparation methods: dehydrohalogenation of alkyl halides; alcohol dehydration, orientation and reactivity of the dehydration; dehalogenation of vicinal dihalides; reduction of alkynes.

3. Reactions of alkenes. Hydrogen addition, addition of hydrogen halides, Markovnikov's rule, Karasch's effect, water addition, ozonolysis and posterior oxidation or reduction, oxidation, hydroxylation, glycol formation. Obtention of thiols and cis-glycols, reaction with hypochlorous acid, oxymercuration-demercuration with water or alcohol, allylic substitution, polymerization.

4. Alkene analysis.
5. Alkynes. Nomenclature. Physical properties. Preparation methods: dehydrohalogenation of vicinal alkyl halides, reaction of sodium acetylides with primary alkyl halides.
6. Reactions of alkynes: addition of hydrogen, halogens, hydrogen halides, addition of water, formation of alkaline and heavy metal acetylides, reaction of acetylene with Grignard, ozonolysis, addition of HClO , polymerization of acetylene.
7. Reactivity of alkenes and alkynes.

Topic VI. Cycloalkanes

1. Baeyer's tension theory. Nomenclature. Six membered ring conformations. Axial and equatorial bonds. Synthesis of cycloalkanes up to 6 carbon atoms. Bicycles. R and S configuration.

Topic VII. Aromatic hydrocarbons

1. Benzene. Aromatic compounds, arenes. Hückel's rule. Resonance energy. Obtention methods. Reactions: nitration, halogenation, Friedel-Crafts' alkylation and acylation, sulfonation.
2. Nomenclature of mono and di-substituted benzene derivatives. Activating and deactivating groups.
3. Fused rings. Naphthalene, anthracene and phenanthrene. Orientation rules.

Topic VIII. Alcohol, ethers and phenols.

1. Alcohols. Structure. Classification and nomenclature. Physical properties.
2. Obtention: Grignard's synthesis and its limitations, hydrolysis of alkyl halides, reduction of carbonylic compounds, reduction of acids and esters, hydroxylation of alkenes, reaction of R-X with wet silver oxide, cleavage of esters with diluted sulfuric acids.
3. Reactions of the alcohols: reaction with hydrogen and phosphorous halides, dehydration, reaction as acids, formation of esters, oxidation, haloform reaction, reduction of alcohols.
4. Identification methods.
5. Ethers. Preparation methods: Williamson's synthesis. Chemical properties: hydrolysis, reaction with concentrated HI and HBr .
6. Phenols. Preparation methods. Chemical properties.

Topic IX. Aldehydes and ketones

1. Geometric and electronic structure. Nomenclature and physical properties.
2. Preparation of aldehydes and ketones: oxidation of alcohols and methylbenzenes, alkene ozonolysis, oxidation of 1,2-glicols with HIO_4 .

3. Reactions: oxidation, haloform reaction, reduction, addition of Grignard's reagent, Clemmensen's reduction, bisulfite addition, Cannizzaro's reaction, addition of ammonia derivatives for identification.

4. Methods of identification of aldehydes and ketones.

Topic X. Carbanions. Condensation reactions

1. Aldolic condensation. Malonic synthesis. Reaction of Hell-Volhard-Zelinski. Acetoacetic synthesis. Claisen's condensation. Dieckmann's condensation. Perkin's reaction. Stobbe's condensation. Michael's reaction. Mannich's reaction. Cope's reaction.

Topic XI. Carboxylic acids

1. Structure. Nomenclature. Physical properties.

2. Preparation of carboxylic acids, oxidation of alkylbenzenes, carbonatation of Grignard's reagents, nitrile hydrolysis, hydrolysis of salts and esters.

3. Chemical reactions, acidity, salt formation, reduction, substitution in the alkyl or aryl group, alpha halogenation in aliphatic acid, Hell-Volhard-Zelinsky's reaction.

4. K_a and substituent effect.

5. Examples of dicarboxylic acids (up to adipic acid).

6. Equivalent of neutralization.

7. Analysis of carboxylic acids and their derivatives: acid halides, amides, anhydrides and esters. Chemical properties.

Topic XII. Amines

1. Structure. Classification. Nomenclature. Physical properties.

2. Methods of obtention: reduction of the nitro or nitrile group, reaction of halides with ammonia.

3. Reactions: halides with ammonia or amines, alcohols aminolysis, electrophilic aromatic substitution in aromatic amines, reactions of amines with nitrous acid, diazonium salts: oreoation and reactions, Sandmeyer's reactions, replacement with OH and with H, replacement with halogen, synthesis with diazonium salts, Hoffman's degradation, Hinsberg's test, Gabriel's synthesis.

Topic XIII. Sulfur compounds

1. Mercaptans or thiols. Nomenclature. Preparation methods. Thioesters. Properties. Thioethers. Preparation methods. Properties.

2. Sulfonic acids. Preparation methods. Properties.

Note: During the classes of Organic Chemistry contents, tools from kinetics, equilibrium, electrochemistry and analytics should be used. Topics I, II and V from tenth grade and topic IV from eleventh grade are included.

Note: When solving many problems, the degree of extension of the reaction may be used.

Programs for the laboratory

10th Grade

1. Demonstrations, class experiments and laboratory practices from units 1-4.
- a) Experimental identification of ions: silver, barium, calcium, aluminum, ammonium, zinc, chromium(III), iron(II) and (III), cobalt(II), nickel(II), copper(II), lead(II), tin(II) and hydrogen ions; chloride, iodides, bromide, sulfide, nitrates, nitrites, sulfate, hydrogencarbonate, carbonate and hydroxyl ions.
- b) Flame identification of ions: sodium, calcium, potassium, barium and strontium.

11th Grade

1. Identification of cations and anions, as stated above.
2. Acid-base titrations.

12th Grade

1. Identification of the main organic functional groups.
2. Methods of laboratory synthesis for alkenes, alcohols, halogenated derivatives, aldehydes, ketones, acids and esters.
3. Objectives from tenth and eleventh grade are included.